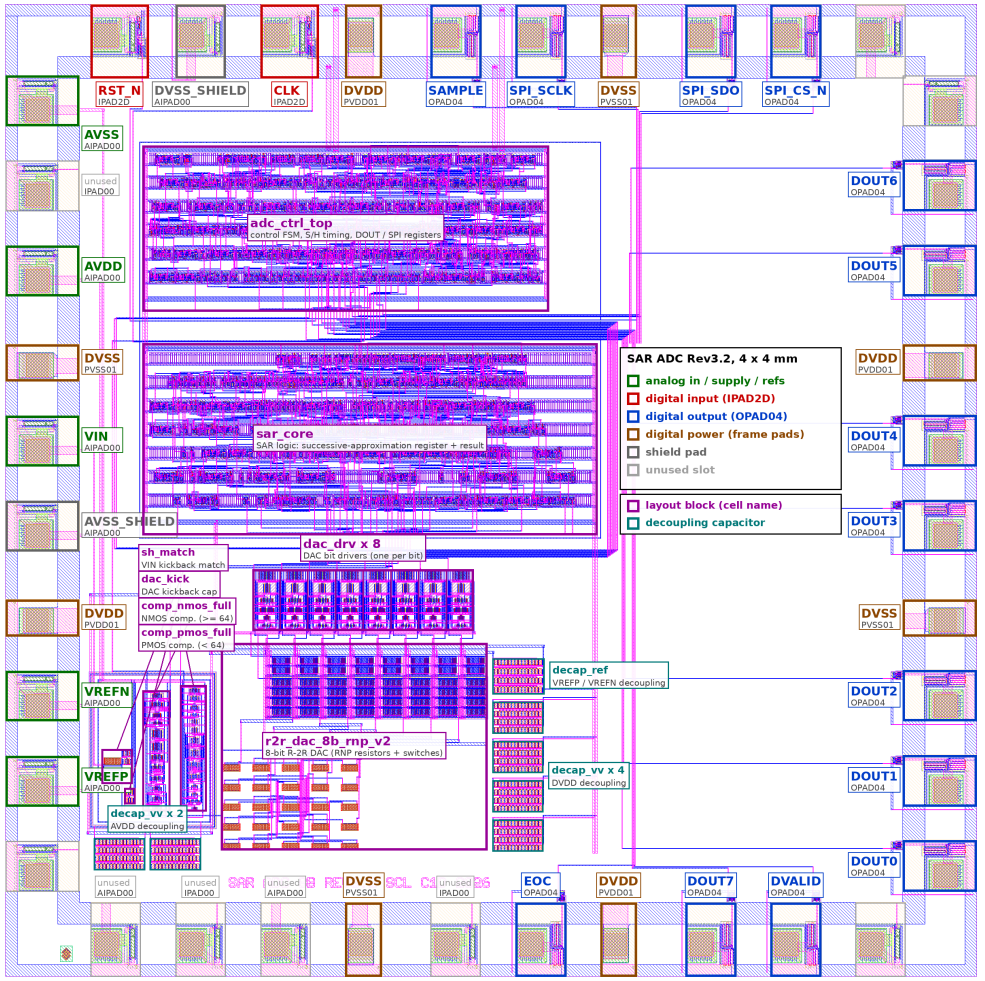
VOICE AI ON INDIAN SILICON

Swar: offline voice AI on a home-grown chip

An 8-bit ADC we designed for India's SCL 1.2 µm fab, paired with keyword AI that runs on the device.

- **Imported ears.** Voice devices depend on imported audio chips.
- **Cloud brains.** Most voice AI needs the internet: latency, privacy and connectivity costs.
- **The gap.** There is no Indian-fab "ear" for voice interfaces.



Our ADC chip: 4 × 4 mm, SCL 1.2 µm, layout verified.

HOW IT WORKS

The chip hears, the FPGA cleans, the AI understands

input
Microphone
Preamp and sample-and-hold on the board.

→

our chip
8-bit SAR ADC
2 MS/s, designed for SCL 1.2 µm.

→

Basys3 FPGA
Oversampling filter
CIC + FIR to 16 kHz audio, about 10-bit quality.

→

ESP32-S3
Keyword AI
Log-mel features and a 21.7k-parameter CNN.

→

output
Action
10 commands, no internet needed.

The trick is oversampling. Sampling 125× faster than the audio rate and filtering on the FPGA turns a cheap 8-bit chip into voice-quality audio: about 10 bits in the voice band.

THE CHIP WE BUILT

A complete ADC, verified ready for fabrication

8-bit
SAR, R-2R DAC

7.8–7.9
ENOB (bits)

≤ 0.27
DNL (LSB), no missing codes

13.9 mW
power at 400 kS/s

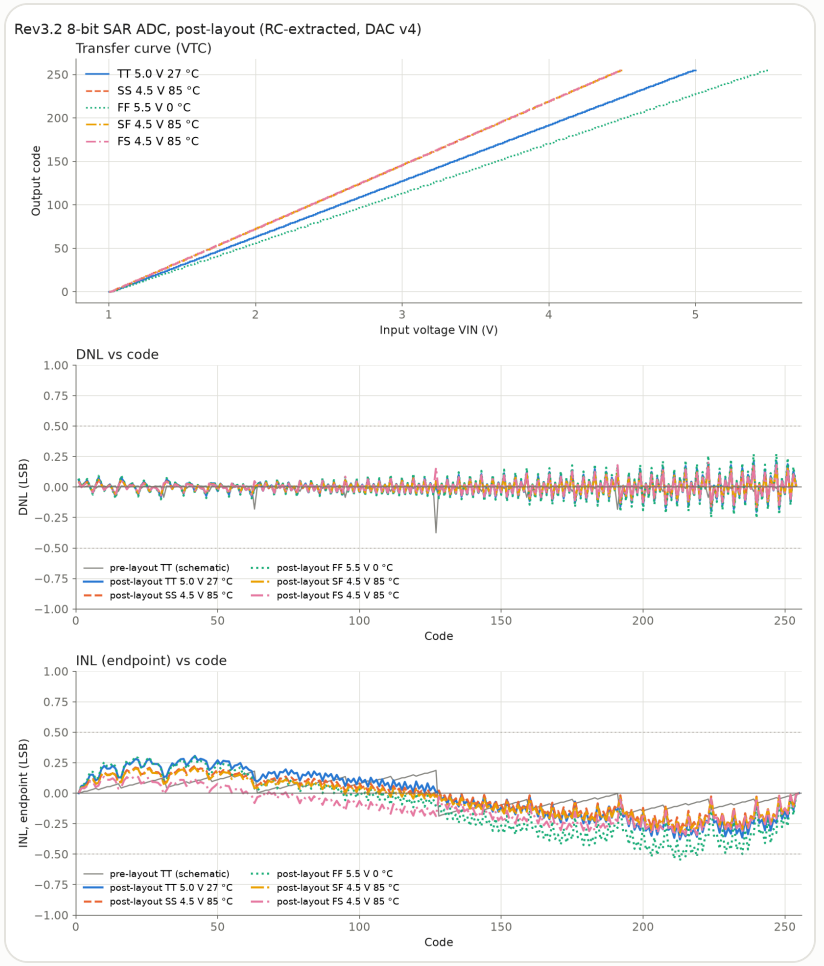
5 V
supply, 4 V input range

4 × 4 mm
die, SCL standard frame

2 MS/s
max sample rate

Clean
DRC 0, LVS match

- **Post-layout verified:** every transistor and wire, at all process corners.
- **Cross-checked:** Cadence Spectre and ngspice agree code for code.



Post-layout transfer curve, DNL and INL at all five process corners.

VOICE AI RESULTS

Keyword accuracy, 10 keywords, 1000 test clips

+16 points over a plain 8-bit converter on quiet speech, within 3 points of a 16-bit mic.



Simulation with the post-layout chip model; silicon pending. Voice-band quality rises from 7.4 to about 10 bits after FPGA oversampling.

IMPACT AND NEXT STEPS

Private, low-cost voice control on Indian silicon

- **Homes and rural areas:** low-cost voice control that works offline.
- **Assistive devices:** hands-free control for the elderly and people with disabilities.
- **Education:** an open voice-AI kit built on an Indian-fab chip.

Code and results: github.com/angelgarg/swar-voice-adc

1. Tape out the chip at SCL.
2. Build the board: mic, sample-and-hold, Basys3, ESP32-S3.
3. Live keyword demo on real silicon.
4. More keywords and Indian languages.
5. Retune the AI on measured silicon data.

Our ask: support for fabrication and board build, and mentorship on deployment.